Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/18/2024
PI Name:	Beheshti, Afshin Ph.D.		
Project Title:	Human Adaptation to Spaceflight: An Integrated Analysis of ISS Data		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Cancer**:Risk of Radiation Carcinogenesis (3) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (4) **Food and Nutrition**:Risk of Performance Decrement and Crew Illness Due to Inadequate Food and Nutrition (5) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response (6) **Renal Stone**:Risk of Renal Stone Formation (7) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS) (8) **Sleep**:Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	beheshti@pitt.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-308-2540
Organization Name:	University of Pittsburgh		
PI Address 1:	4200 Fifth Ave		
PI Address 2:			
PI Web Page:			
City:	Pittsburgh	**State:**	PA
Zip Code:	15260	**Congressional District:**	12
Comments:	PI has moved from NASA Ames Research Center to the University of Pittsburgh, where he serves as Director of the Center for Space Biomedicine. Former email: afshin.beheshti@nasa.gov		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	02/01/2024	**End Date:**	08/01/2027
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			

COI Name (Institution):	Costes, Sylvain Ph.D. (NASA Ames Research Center) Dragsted, Lars Ph.D. (University of Copenhagen) Metz, Gerlinde M.D., Ph.D. (The University of Lethbridge) olde Scheper, Tjeerd Ph.D. (Oxford Brookes University) Smith, Scott Ph.D. (NASA Johnson Space Center) Fazelinia, Hossein Ph.D. (Children's Hospital of Philadelphia) Galeano, Diego Ph.D. (Universidad Nacional de Asunción) Laiakis, Evagelia Ph.D. (Georgetown University) McDonald, John Ph.D. (SRI International) Schisler, Jonathan Ph.D. (University of North Carolina, Chapel Hill) Zwart, Sara Ph.D. (NASA Johnson Space Center) Wallace, Douglas M.D., Ph.D. (Children's Hospital of Philadelphia)
Grant/Contract No.:	Directed Research
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The systemic effects of prolonged space missions on various organs are a significant concern, and a knowledge gap exists regarding the crucial pathways and factors contributing to associated health risks. Our focus centers on mitochondrial dysfunction as a central hub affected by the space environment, leading to immune dysregulation. Analysis of samples from astronauts and mice after extended stays on the International Space Station (ISS) highlights adverse effects on the mitochondrial energy system, oxidative phosphorylation (OXPHOS). The recent National Academies of Sciences, Engineering, and Medicine (NASEM) Decadal Survey underscores the importance of mitochondria and metabolic research in the next decade of space biology. Mitochondrial suppression, particularly in OXPHOS, poses risks to cognitive abilities and vision due to its role in generating 90% of the body's energy. Maternally inherited mitochondrial DNA (mtDNA) is especially impacted during space exposure, reducing cell and brain energy capacity. We hypothesize that spaceflight induces systemic dysregulation of mitochondrial activity, affecting immune response, Spaceflight Associated Neuro-Ocular Syndrome (SANS), metabolism, and bone health. Our systems biology approach aims to characterize systemic metabolic changes in astronauts and identify potential countermeasures using machine learning techniques with the following specific aims: 1) Assess individual metabolic and environmental factors related to SANS outcomes, specifically increased total retinal thickness postflight. 2) Investigate sex differences in metabolic responses to spaceflight, considering dietary intake, exercise, and environmental impact on body mass and composition. 3) Examine individual and sex differences in biochemical, endocrine, immune function, and omics measures in relation to dietary, exercise, and environmental factors affecting bone loss. 4) Identify dysregulated circulating mitochondrial factors in spaceflight, contributing to increased health risks related to reduced bioenergetics. Our scientific team will analyze extensive astronaut data, including biochemistry, endocrinology, cytokine profiles, metabolomics, proteomics, and lipidomics. Physiological data on body mass, bone and body composition, SANS status, exercise, medication logs, and dietary intake will be considered. Environmental data, such as CO2, temperature, humidity, and radiation dosimetry, will be correlated with biological changes. This multi-omics, multi-level, systems biology approach is the first on a comprehensive ISS astronaut population. Lastly, novel machine learning techniques will predict FDA-approved drugs for repurposing as countermeasures to mitigate determined metabolic changes.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research.
Research Impact/Earth Benefits:	This project will allow for the determination of key systemic circulating factors that will impact different aspects of key metabolic activity changing directly in astronauts, contributing to increased health risks. It will be the largest omics dataset from astronauts to be analyzed, which will provide both novel findings and confirmation for previous studies on other experimental models (i.e., in vitro and in vivo models). In addition, we will be making the deidentified data available to the public on NASA’s Open Science Data Repository (OSDR), which provides an enormous impact to the scientific community to produce future insight and opportunities.
Task Progress:	New Project for FY2024
Bibliography Type:	Description: (Last Updated: 07/13/2026)